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# -----
banner("STEP 5 (bonus) – Causal models (GPT-2) only see the LEFT context")
# -----
# In both sentences the words BEFORE "bank" are identical; the disambiguating words
# come AFTER. A bidirectional model can use them, a causal (decoder-only) model cannot.
s1 = "The bank was flooded by the river."
s2 = "The bank was robbed by two men."
print(f"  A: {s1}\n    B: {s2}\n")

for name in ["bert-base-uncased", "gpt2"]:
    tok = AutoTokenizer.from_pretrained(name)
    mdl = AutoModel.from_pretrained(name, output_hidden_states=True).eval()
    va, ta = target_vectors(s1, TARGET, tok, mdl)
    vb, _ = target_vectors(s2, TARGET, tok, mdl)
    kind = "bidirectional" if "bert" in name else "causal (left-to-right)"
    print(f"  {name:18s} [{kind:22s}] cosine('{TARGET}' in A, in B) at last layer = "
          f"{cos(va[-1], vb[-1]):.4f}")
print("\n  GPT-2 gives exactly 1.0000: when it reads 'bank' it has not seen 'river' or 'robbed' yet."
      "\n  The meaning is resolved later, in the vectors of the FOLLOWING tokens."
      "\n  (Extra caveat for GPT-2: its last-layer vectors are very anisotropic -- almost all"
      "\n  pairs have high cosine similarity -- so compare differences, not absolute values.)")

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